Prince Benedict Wachira

Full stack & Devopps Engineer

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SUMMARY

Full Stack Software Engineer with strong expertise in backend engineering and DevOps, experienced in designing, building, deploying, and maintaining scalable cloud-native applications from development to production. Proficient in **Node.js, Express.js, Go, PostgreSQL, MongoDB, React, and React Native**, with hands-on experience architecting RESTful APIs, database systems, authentication, and distributed applications. Skilled in modern DevOps practices, including **Kubernetes, Docker, GitHub Actions, ArgoCD, Velero, AWS S3, Linux administration, CI/CD automation, monitoring, and disaster recovery**, enabling reliable, secure, and highly available production environments. Passionate about automating infrastructure, optimizing application performance, and delivering end-to-end software solutions that scale. Currently pursuing a **Bachelor's degree in Applied Computer Science** while advancing backend engineering expertise through the **ALX Backend Developer Program**.

PROFESSIONAL EXPERIENCE

Loomlights Limited , *Backend & DevOps Engineer* 07/2026 | Nairobi

- Designed, developed, and maintained backend services and RESTful APIs for the **TapAndGo** water tracking platform.
- Architected, deployed, and managed production **Kubernetes (K3s)** infrastructure hosting multiple applications.
- Built and maintained automated **CI/CD pipelines** using **GitHub Actions** and **ArgoCD**, enabling reliable GitOps-based deployments.
- Automated Docker image building, testing, and deployment to Kubernetes clusters with rolling updates and rollback strategies.
- Configured **Longhorn** distributed storage for Kubernetes persistent volumes and high-availability workloads.
- Implemented automated backup and disaster recovery solutions using **Velero** with **AWS S3** for secure backup storage.
- Managed application assets, object storage, and backup infrastructure using **AWS S3**.
- Configured centralized monitoring, logging, and observability using **Grafana, Prometheus, Loki, and Uptime Kuma**, improving production visibility and incident response.
- Managed **Traefik** Ingress Controller, DNS routing, TLS certificates, and production networking for secure application delivery.
- Administered Ubuntu Linux servers and production infrastructure while troubleshooting and resolving production incidents to improve system reliability and uptime.
- Collaborated closely with software engineers to streamline deployment workflows, infrastructure automation, and application releases.
- **Designed and developed three production web applications:** the **Loomlights** corporate website, the **TapAndGo** platform, and the **RecycleRush** platform, delivering responsive user interfaces integrated with scalable backend services.

Opencrafts Limited, *Backend and Web Contributor* 05/2025 | Nairobi

- Designed and developed scalable backend services for the **Academia** platform using **Node.js, Express.js, and PostgreSQL**.
- Architected and implemented the backend for **Sherehe**, an event management system, including RESTful APIs, authentication, ticketing, and database design.

- Designed and built a complete **Bus Booking System**, implementing trip scheduling, seat management, booking workflows, payment integration, and reservation APIs.
- Built and maintained RESTful APIs following best practices for scalability, security, and maintainability.
- Designed relational database schemas and optimized PostgreSQL queries to improve application performance and data reliability.
- Collaborated with frontend developers to integrate backend services and deliver seamless user experiences.
- Improved API performance through query optimization, efficient data modeling, and backend refactoring.
- Contributed to the organization's open-source software ecosystem by developing and maintaining backend components across multiple projects.
- Participated in code reviews, feature planning, debugging, and deployment to ensure high-quality software delivery.

Craygroup Limited 🔗 , *Full stack $ DevOps Engineer* 09/2024 | Nairobi

- Designed and developed a complete **ISP Ticketing & Customer Support System** using **Node.js, Express.js, PostgreSQL, Sequelize, and React Native**, providing ticket management, customer support, and field staff workflows.
- Implemented **SLA enforcement**, ticket workflow automation, role-based access control (RBAC), audit logging, and ticket history tracking.
- Integrated subscriber management with **Hotspot** and **PPPoE** systems to streamline ISP customer operations.
- Developed staff performance analytics, reporting dashboards, and operational insights to improve support efficiency.
- Contributed to the development of the **Aetherlink** web application, building backend services and features for managing ISP customers, subscriptions, and user accounts.
- Designed, deployed, and maintained a production **Kubernetes (K3s)** infrastructure hosting multiple business applications.
- Built and managed a **multi-node Kubernetes cluster** by connecting multiple Ubuntu Linux servers into a unified production environment.
- Implemented **GitOps** deployment workflows using **ArgoCD** for automated Kubernetes application delivery.
- Designed and maintained **CI/CD pipelines** using **GitHub Actions**, optimizing workflows with **Blacksmith** to reduce build times.
- Automated Android application build, signing, and release workflows using **Fastlane**.
- Automated Docker image building, testing, and deployments to Kubernetes clusters.
- Configured **Longhorn** distributed storage and implemented automated backup and disaster recovery using **Velero** with **AWS S3-compatible object storage**, including off-site Longhorn backups.
- Managed application assets and backup storage using **AWS S3**.
- Configured centralized monitoring and observability using **Grafana, Prometheus, Loki, and Uptime Kuma**, improving system visibility and incident response.
- Managed **Traefik Ingress Controller**, DNS routing, HTTPS, TLS certificates, and production networking.
- Administered Ubuntu Linux servers, storage systems, networking, and production infrastructure.
- Integrated **Firebase Cloud Messaging (FCM)** for push notifications, **Resend** for transactional email delivery, and **Umami Analytics** for privacy-focused application monitoring.
- Investigated and resolved production incidents involving Kubernetes, networking, storage, and application infrastructure, improving overall reliability and availability.
- Authored technical documentation, deployment guides, operational runbooks, incident reports, and post-mortem analyses to support knowledge sharing and continuous improvement.
- Collaborated closely with software engineers to automate deployments, improve release processes, and maintain scalable, secure, and highly available production environments.

EDUCATION

Daystar University, *BSc in Applied Computer Science* 09/2023 – 09/2026

- Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Operating Systems, Computer Networks, Web Development, Mobile Application Development, Artificial Intelligence, Cybersecurity, and Distributed Systems.

ALX Africa, *Backend Software Engineering Program* 07/2025 – Present

- Intensive software engineering program focused on backend development, scalable system design, RESTful APIs, databases, Linux, DevOps, cloud technologies, and professional software engineering practices.
- Developed hands-on experience building production-ready backend applications using modern development workflows and industry best practices.

TECHNICAL SKILLS

Programming Languages (JavaScript (ES6+) | Go | SQL)

Frontend Development (React | React Native | HTML5 | CSS3 | Responsive Web Design)

Backend Development (Node.js | Golang | Express.js | RESTful API Development | Authentication & Authorization | Microservices | Webhooks | RabbitMQ)

Database Technologies (PostgreSQL | MongoDB | Sequelize ORM | Database Design | Query Optimization | Schema Migrations | Data Modeling)

DevOps & Cloud (Kubernetes (K3s) | Docker | GitHub Actions | ArgoCD (GitOps) | AWS S3 | Velero | Longhorn | Traefik | Linux (Ubuntu Server))

CI/CD & Release Engineering (Continuous Integration & Continuous Delivery (CI/CD) | GitHub Actions | GitOps Workflows | Release Automation | Automated Build & Deployment Pipelines | Blacksmith CI Acceleration | Fastlane (Android Build & Release))

Monitoring & Observability (Grafana | Prometheus | Loki | Uptime Kuma | Production Monitoring | Log Aggregation | Incident Response)

Mobile Development (React Native | Expo | Android Studio | Firebase Cloud Messaging (FCM))

Version Control & Collaboration (Git | GitHub | Agile Development | Code Reviews)

Documentation & Operations (Technical Documentation | Infrastructure Documentation | Deployment Guides | Operational Runbooks | Incident Reporting | Root Cause Analysis (RCA) | Production Support | Change Management)

Integrations & Developer Tools (Resend | Umami Analytics | Postman | Figma | VS Code)

TOOLS & TECHNOLOGIES

Cloud & Infrastructure
Kubernetes (K3s) , Docker , Longhorn , Velero , AWS S3 , Traefik , Cert-Manager , Contabo Cloud , Ubuntu Linux , Bash , SSH

CI/CD & DevOps
GitHub Actions , ArgoCD , Blacksmith , Fastlane , Git ,GitHub

Monitoring & Observability
Grafana , Prometheus , Loki , Uptime Kuma , Umami Analytics

Backend & Databases
Node.js , Express.js , PostgreSQL , MongoDB , MySQL , Sequelize ORM , RabbitMQ , REST APIs

Development Tools
Visual Studio Code , Postman , Insomnia , Figma

Messaging & Notifications
Firebase Cloud Messaging (FCM) , Resend , Mailcow

SOFT SKILLS

- Infrastructure Automation
- Problem Solving
- Analytical Thinking
- Troubleshooting & Incident Resolution
- System Reliability & Availability
- Team Collaboration
- Communication
- Technical Documentation
- Incident Reporting & Root Cause Analysis
- Time Management
- Adaptability
- Continuous Learning
- Attention to Detail
- Production Support
- DevOps Collaboration (Dev & Ops)

PROJECTS

Cloud Infrastructure & Kubernetes Platform, *Designed, deployed, and managed production-grade cloud-native infrastructure supporting multiple business applications.* 06/2024 – Present
Key Achievements
- Designed and deployed a production Kubernetes (K3s) cluster across multiple Linux servers.
- Configured and managed Longhorn distributed storage for persistent Kubernetes workloads.
- Implemented automated backup and disaster recovery using Velero with AWS S3-compatible object storage.
- Configured Longhorn backups to an off-site storage server outside the production network.
- Managed Traefik Ingress Controller, DNS routing, TLS certificates, and HTTPS configuration.
- Configured production networking and service discovery for Kubernetes applications.
- Administered Ubuntu Linux servers and production environments.
- Improved infrastructure reliability through monitoring, automation, and disaster recovery planning.

Technologies: Kubernetes (K3s), Docker, Longhorn, Velero, AWS S3, Traefik, Ubuntu Linux

CI/CD & GitOps Automation Platform, *Designed and implemented automated software delivery pipelines for containerized applications.* 07/2024 – Present
Key Achievements
- Built CI/CD pipelines using GitHub Actions.
- Implemented GitOps deployment workflows using ArgoCD.
- Automated Docker image building, testing, and deployment.
- Integrated Blacksmith to accelerate GitHub Actions workflows and reduce build times.

- Implemented Fastlane to automate Android application build, signing, and release processes.
- Standardized deployment workflows across multiple applications.
- Reduced deployment time through automation and repeatable release processes.

Technologies: GitHub Actions, ArgoCD, Docker, Blacksmith, Fastlane

Monitoring & Observability Platform 07/2025 – Present

Designed a centralized monitoring and observability platform for production systems.

Key Achievements

- Implemented Grafana dashboards for infrastructure monitoring.
- Configured Prometheus for metrics collection.
- Centralized application and infrastructure logs using Loki.
- Implemented uptime monitoring and alerting with Uptime Kuma.
- Improved incident detection and troubleshooting through proactive monitoring.

Technologies: Grafana, Prometheus, Loki, Uptime Kuma

Notification & Communication Platform 01/2026 – Present

Designed and integrated reliable notification services for web and mobile applications.

Key Achievements

- Integrated Firebase Cloud Messaging (FCM) for push notifications.
- Integrated Resend for transactional email delivery.
- Built a self-hosted Mailcow email infrastructure.
- Implemented reusable email templates.
- Designed queue-based email processing.
- Improved notification reliability and delivery performance.

Technologies: Firebase Cloud Messaging, Resend, Mailcow, Node.js, REST APIs

Analytics & User Insights Platform, *Implemented privacy-focused analytics for production applications.*

Key Achievements

- Deployed and configured Umami Analytics.
- Implemented event tracking across web applications.
- Integrated privacy-friendly analytics dashboards.
- Improved visibility into user engagement and application performance.

Technologies: Umami Analytics, Docker, PostgreSQL

Schemos (Database Migration Framework) , *Sequelize-native migration tool - automatically generates and applies migrations from your Sequelize models"* 07/2026 – Present

- Designed and developed **Schemos**, a PostgreSQL schema migration framework that automatically compares Sequelize models with live database schemas and generates safe SQL migrations

Key achievements

- Built an automatic schema comparison engine, eliminating the need to manually write migration scripts.
- Developed a migration generator capable of detecting new tables, columns, indexes, constraints, and relationships.
- Implemented SQL preview functionality, allowing developers to review migrations before execution and reducing deployment risks.
- Created a migration runner that executes migrations safely while maintaining schema consistency across environments.
- Reduced database deployment time by automating repetitive migration tasks and minimizing human error.

*Technologies***:** Node.js, PostgreSQL, Sequelize

Email Delivery Platform 08/2025 – Present

Developed a self-hosted email delivery service similar to Resend, providing businesses with an affordable API for transactional emails.

Key achievements

- Built REST endpoints that allow applications to send emails programmatically.
- Integrated Mailcow infrastructure to provide reliable SMTP delivery.
- Designed reusable email templates and queue-based email processing.
- Reduced dependency on third-party email providers by offering a lower-cost internal solution.
- Built the platform to support future features including delivery tracking and analytics.

Technologies: Node.js, Mailcow, REST APIs

ISP Ticketing & Support Platform 04/2026 – Present

Designed and developed the complete backend for an ISP ticketing platform supporting customer service operations and staff management.

Key achievements

- Built a scalable REST API serving ticket management, subscriber management, and staff operations.
- Implemented SLA monitoring and automated ticket escalation, improving response time tracking.
- Designed a role-based access control (RBAC) system to secure administrative and customer operations.
- Integrated Hotspot and PPPoE subscriber information into the support workflow, enabling agents to troubleshoot customers directly from the platform.
- Developed staff performance analytics that automatically calculate assignment, response, and resolution metrics.
- Implemented comprehensive audit logging for every ticket update, improving accountability and troubleshooting.

Technologies: Node.js, Express.js, PostgreSQL, Sequelize

Event Management Backend (Sherehe) 06/2025 – Present

Built the backend infrastructure for an event management platform supporting organizers, attendees, and ticket verification.

Key achievements

- Designed REST APIs for event creation, attendee registration, ticket purchases, and event management.
- Built QR-code ticket generation and validation services to reduce fraudulent ticket usage.
- Implemented email notification workflows for registrations, invitations, and ticket confirmations.
- Developed a scalable PostgreSQL database architecture supporting thousands of ticket records.
- Created authentication and authorization systems to separate organizer and attendee privileges.

Technologies: Node.js, Express.js, PostgreSQL, RabbitMQ

Email Sender (Resend-like Platform)

Developed a self-hosted email delivery service similar to Resend, providing businesses with an affordable API for transactional emails.

Key achievements

- Built REST endpoints that allow applications to send emails programmatically.
- Integrated Mailcow infrastructure to provide reliable SMTP delivery.
- Designed reusable email templates and queue-based email processing.
- Reduced dependency on third-party email providers by offering a lower-cost internal solution.
- Built the platform to support future features including delivery tracking and analytics.

Technologies: Node.js, Mailcow, REST APIs

VOLUNTEER ACTIVITIES

Daystar Information Technology Association (DITA) — Backend & Web Development Contributor | 2024 – Present • Contributed to the development and maintenance of the official DITA website. • Collaborated with fellow student developers to implement new features and improve system functionality. • Participated in testing, debugging, and optimizing web application performance. | **Daystar University Chapel — Volunteer Developer – Chapel Attendance System** | 2025 • Contributed to the development of a digital chapel attendance management system. • Assisted in implementing backend functionality for attendance recording and reporting. • Helped modernize attendance tracking by replacing manual processes with a digital solution. | **Kenya Red Cross Society — Active Member & Volunteer Blood Donor** | Ongoing • Active member of the Kenya Red Cross Society, participating in community service and humanitarian initiatives. • Regular voluntary blood donor, supporting national efforts to maintain safe and adequate blood supplies. • Participate in outreach activities that promote health awareness, emergency preparedness, and community resilience. | **Hospice Care Volunteer — Volunteer** | Ongoing • Support hospice initiatives that promote compassionate care and improve the quality of life for patients who have terminal illness and their families. • Participate in community outreach and volunteer activities organized by the hospice. • Demonstrate empathy, teamwork, and a commitment to serving vulnerable members of the community.